Schema design

1. **MEMBER (member_id PK)**

├── plan_id → MembershipPlan(plan_id)

├── trainer_id → Trainer(trainer_id)

```
mysql> desc member;
+------------+--------------+------+-----+---------+----------------+
| Field      | Type         | Null | Key | Default | Extra          |
+------------+--------------+------+-----+---------+----------------+
| member_id  | int          | NO   | PRI | NULL    | auto_increment |
| name       | varchar(100) | YES  |     | NULL    |                |
| age        | int          | YES  |     | NULL    |                |
| gender     | varchar(10)  | YES  |     | NULL    |                |
| phone      | varchar(15)  | YES  |     | NULL    |                |
| join_date  | date         | YES  |     | NULL    |                |
| plan_id    | int          | YES  | MUL | NULL    |                |
| trainer_id | int          | YES  | MUL | NULL    |                |
| password   | varchar(100) | YES  |     | NULL    |                |
+------------+--------------+------+-----+---------+----------------+
9 rows in set (0.09 sec)
```

2. **TRAINER (trainer_id PK)**

```
mysql> desc trainer;
+----------------+--------------+------+-----+---------+----------------+
| Field          | Type         | Null | Key | Default | Extra          |
+----------------+--------------+------+-----+---------+----------------+
| trainer_id     | int          | NO   | PRI | NULL    | auto_increment |
| name           | varchar(100) | YES  |     | NULL    |                |
| specialization | varchar(100) | YES  |     | NULL    |                |
| phone          | int          | YES  |     | NULL    |                |
| password       | varchar(20)  | YES  |     | NULL    |                |
+----------------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

3. **WORKOUTPLAN (workout_id PK)**

├── member_id → Member(member_id)

├── trainer_id → Trainer(trainer_id)

```
mysql> desc workoutplan;
+------------+------+------+-----+---------+----------------+
| Field      | Type | Null | Key | Default | Extra          |
+------------+------+------+-----+---------+----------------+
| workout_id | int  | NO   | PRI | NULL    | auto_increment |
| member_id  | int  | YES  | MUL | NULL    |                |
| trainer_id | int  | YES  | MUL | NULL    |                |
| exercises  | text | YES  |     | NULL    |                |
| date       | date | YES  |     | NULL    |                |
+------------+------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

4. **PAYMENT (payment_id PK)**

├── member_id → Member(member_id)

```
mysql> desc payment;
+------------+---------------+------+-----+---------+----------------+
| Field      | Type          | Null | Key | Default | Extra          |
+------------+---------------+------+-----+---------+----------------+
| payment_id | int           | NO   | PRI | NULL    | auto_increment |
| member_id  | int           | YES  | MUL | NULL    |                |
| amount     | decimal(10,2) | YES  |     | NULL    |                |
| date       | date          | YES  |     | NULL    |                |
| status     | varchar(20)   | YES  |     | NULL    |                |
+------------+---------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

5. **ATTENDANCE (attendance_id PK)**

├── member_id → Member(member_id)

```
mysql> desc attendance;
+---------------+------+------+-----+---------+----------------+
| Field         | Type | Null | Key | Default | Extra          |
+---------------+------+------+-----+---------+----------------+
| attendance_id | int  | NO   | PRI | NULL    | auto_increment |
| member_id     | int  | YES  | MUL | NULL    |                |
| date          | date | YES  |     | NULL    |                |
+---------------+------+------+-----+---------+----------------+
```

6. **MEMBERSHIPPLAN (plan_id PK)**

```
mysql> desc membershipplan;
+----------+---------------+------+-----+---------+----------------+
| Field    | Type          | Null | Key | Default | Extra          |
+----------+---------------+------+-----+---------+----------------+
| plan_id  | int           | NO   | PRI | NULL    | auto_increment |
| type     | varchar(50)   | YES  |     | NULL    |                |
| price    | decimal(10,2) | YES  |     | NULL    |                |
| duration | int           | YES  |     | NULL    |                |
+----------+---------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)
```

The database consists of six main tables: Member, Trainer, WorkoutPlan, MembershipPlan, Payment, and Attendance. These tables are interconnected using primary and foreign key relationships to ensure data integrity.